1. Introduction

This document is an original teaching fixture for Lemma.
It contains a deliberately false statement and a counterexample.
No result in this document is presented as new research.

Read page 3 to inspect the annotated proposition.

2. Definitions and scope

A sequence f_n converges pointwise to f when:
for each x and each epsilon > 0, there exists N(x, epsilon)
such that |f_n(x) - f(x)| < epsilon whenever n >= N.

Uniform convergence requires N to be independent of x.

2. Limits of continuous functions

In this section we examine pointwise limits and continuity.
The proposition below is deliberately false: it is included
as a teaching example for a paper-review workbench.

Proposition 2.1. *The pointwise limit of continuous functions on [0,1] is continuous.*

A proposed argument. Let f_n be continuous and let f_n tend
to f pointwise. For each x and each epsilon > 0, there is an
N = N(x, epsilon) such that the difference is small for n >= N.
Choosing a single N for every x would require justification.

Example 2.2. For n >= 1, consider the continuous functions

$$f_n(x) = x^n, \qquad x \in [0,1].$$

For 0 <= x < 1 the limit is 0; at x = 1 the limit is 1.
Thus the pointwise limit f is given by

$$f(x) = 0 \quad \text{if } 0 \le x < 1; \qquad f(1) = 1.$$

The left-hand limit of f at 1 is 0, but f(1) = 1.
Consequently, f is not continuous at 1.

Remark 2.3. A corrected statement uses uniform convergence.

The uniform limit of continuous functions is continuous.
This is a different theorem; see the next page for its proof.

4. Uniform convergence

Theorem 3.1. Let X be a metric space. If continuous functions f_n : X -> R converge uniformly to f, then f is continuous.

Unlike Proposition 2.1, this statement assumes uniform convergence. The change in hypothesis must remain explicit.

5. Proof of the corrected statement

Fix a point x_0 and epsilon > 0. By uniform convergence,
choose n such that |f_n(x)-f(x)| < epsilon/3 for every x.
By continuity of f_n at x_0, choose delta > 0 with
|f_n(x)-f_n(x_0)| < epsilon/3 if d(x,x_0) < delta.
The triangle inequality gives |f(x)-f(x_0)| < epsilon.

This argument verifies the corrected theorem informally.
It is not a machine-checked proof.

6. Examples and non-examples

Example: f_n(x) = x/n on [0,1] converges uniformly to 0. Non-example: f_n(x) = x^n on [0,1] is only pointwise convergent; the limit has a discontinuity at the endpoint.

7. References and dependencies

This teaching fixture is self-authored.
Formal statements, external citations and their hypotheses
must be checked against their original sources in real reviews.

8. Review notes

Review scope: teaching proposition and the quantifier issue.
Agent handoffs shown in the application are demonstration data.
Lean has not been run; there is no proof certificate.

User-imported papers start with no findings.